

GRAILCARD



Photograph a card. Know what it is, and what it's worth.

The system identifies the exact printing, measures the card's condition against the professional grading standard, and values it from real completed sales — in Australian dollars.

TIME PER CARD	CONDITION	VALUATION	COVERAGE
About 45 seconds	Measured to PSA tolerance	From completed sales, in AUD	8 card games

Three questions, answered in order

Each stage depends on the one before it. Getting the printing wrong makes every number after it meaningless, so identification comes first.

STAGE ONE

Which card is this?

Down to the set and the collector number. Two cards can share a name and differ in value a thousandfold, so this is settled before anything else.

STAGE TWO

What condition is it in?

Centering measured from the printed border and compared to the PSA standard. Surface marks found and shown on the card itself.

STAGE THREE

What is it worth?

Graded value if it is in a slab, raw value if it is not — taken from real completed sales and converted to Australian dollars.

THE COMMERCIAL POINT

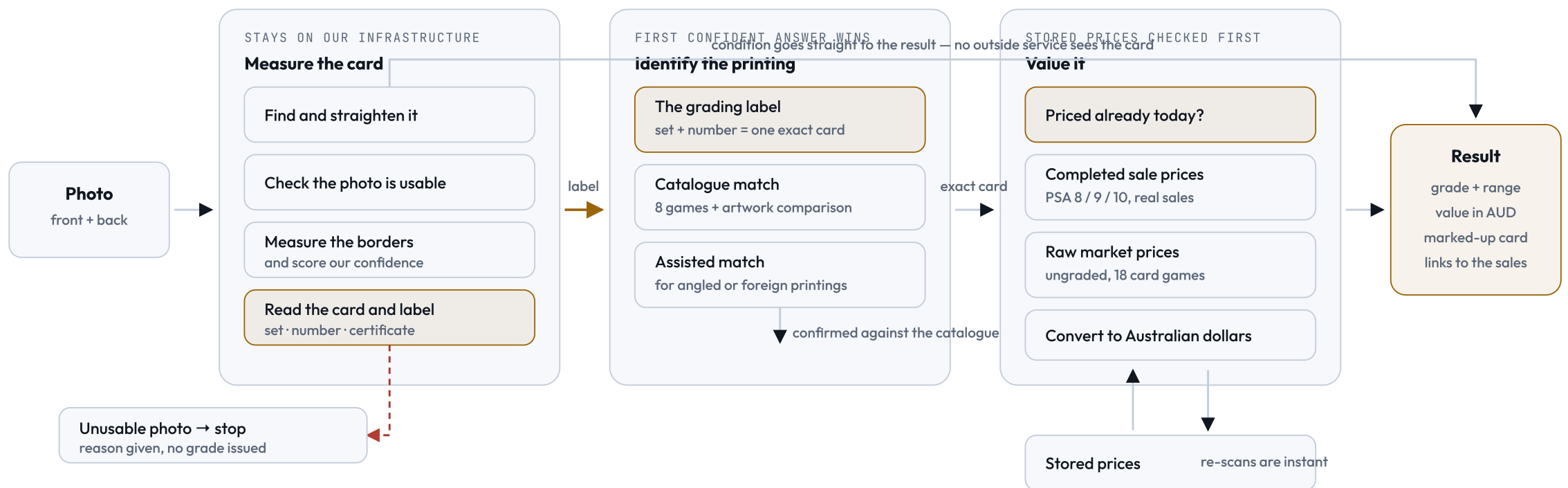
A card in a grading slab is worth its **graded** price. Most price guides quote the raw, ungraded figure. For the card on slide O6 that difference is A\$5,580 against A\$19,949 — the same object, described two ways.

WHAT IT DOES NOT CLAIM

The system identifies a card; it does not authenticate one. It can tell you which card this is meant to be, not that it is genuine. High-value cards should still go to a grading company.

How the pieces fit together

Condition is measured on our own infrastructure and never leaves it. Only the card's identity and its price reach outside, and both are stored so the same card is never looked up twice.



 the high-confidence path — a graded card identifies itself

--- early stop — the system declines rather than guesses

The gold path is the one tried first on every scan. The alternatives exist because photographs of real cards, taken on real phones, frequently do not cooperate.

How the grading actually works

Centering is a measurement, not an opinion, and this is the sequence that produces it.

1 Find the card and correct the angle

A card is 63×88 mm, so anything whose shape falls outside a sensible window around that ratio is a table edge or a sleeve, not the card. Once found, the perspective is corrected onto a fixed canvas — a card photographed at an angle then measures the same as one shot straight down.

2 Decide whether the photo can be trusted at all

Focus is scored, glare is measured as a percentage of the card surface, and the available detail is checked. Failing any one stops the scan with a plain reason and a re-shoot instruction, and no grade is issued.

glare above 4% of the surface • not enough detail • out of focus

3 Walk inward from each edge to find the printed border

Starting at the card edge, the colour is followed pixel by pixel inward. The moment it departs from the border colour and stays departed, the print edge has been found. Only the middle of each side is sampled, so rounded corners cannot skew the result. Four edges give four distances, and those give the centering.

sampled across the central 15-85% of each side

4 Score the measurement before reporting it

A reading is only as good as its evidence. The border ring is checked for uniformity and real contrast. Where a card has no printed border, or the lighting has destroyed it, the result is marked unmeasurable rather than published as a number — and a confidence percentage travels with every figure that is published.

confidence published 0-100% alongside the measurement

5 Compare to the professional standard and show the marks

Three ways to name a card, best first

The system stops at the first method that gives a confident answer, so the most reliable evidence always wins.

1 Read the grading label

A graded card carries its own answer key. The label prints the year, the set and the collector number, and those three together resolve to exactly one card — no name matching and no interpretation. This is tried first on every scan.

2 Match the card itself against the catalogues

For raw cards, the text on the card is read and eight game catalogues are searched at once. The photograph is then compared against each catalogue's own artwork to settle which printing it is.

3 Fall back to an assisted match — then confirm it

For steep angles, heavy glare or Japanese printings, an image model proposes a name. That proposal is never shown on its own: it goes back through the real catalogue, and only a confirmed catalogue entry reaches the screen.

Where none of the three produces a confident answer, the system reports what it read off the card and shows no price — rather than attaching a plausible figure from the wrong set.

Graded and raw are different markets

A real card the system valued — a Charizard ☆ δ from EX Dragon Frontiers, certified BGS 8.5.
All figures in Australian dollars.

Raw and ungraded — the figure most price guides quote	A\$5,580
In its BGS 8.5 slab — what the card is actually worth	A\$19,949
The same card at PSA 9	A\$21,101
The same card at PSA 10	A\$81,924

Quoting the raw figure for this card understates it by roughly three and a half times. Recognising that the card is slabbed, reading which grade it holds, and pricing that grade is the difference between a useful valuation and a misleading one.

PROVENANCE ON EVERY FIGURE

- VERIFIED SALES
- ESTIMATED

Green means real completed sales. Amber means a calculated figure. Every number carries one of the two.

CURRENCY

Australian dollars by default, converted at daily central-bank reference rates. The original figure stays visible so nothing is hidden behind the conversion.

CHECKABLE

Every result links straight out to the completed sold listings, so the figure can be verified against the open market in one click.

Every valuation can show its working

The A\$19,949 on the previous slide is not an estimate. It is the median of real completed sales, and the sample behind it travels with the figure.

PSA 8 · THE TIER THIS BGS 8.5 SITS IN

29 SALES

A\$19,949

A\$11,719

median

A\$27,902

Twenty-nine completed sales. The median is published rather than the average, so a single outlier auction cannot move a valuation.

PSA 10 · SAME CARD, TOP GRADE

4 SALES ONLY

A\$81,924

A\$43,248

A\$90,682

Four sales across a wide spread. A thin sample is itself information, so the count and the range are shown alongside the number rather than hidden behind it.

Where no completed sales exist for a card, the system shows the raw market price and says so. It does not manufacture a graded figure to fill the space.

Actual output, unedited

Certificate numbers, measurements and valuations exactly as the running system produced them.

CARD	READ FROM THE LABEL	CENTERING MEASURED	VALUE RETURNED
Charizard ☆ δ EX Dragon Frontiers #100	BGS 8.5 cert 0011755115	50 / 50 · 56 / 44 passes PSA 10 · 100% confidence	A\$19,949 VERIFIED SALES
Umbreon VMAX Evolving Skies #215 · alternate art secret	PSA GEM MT 10 cert 85183760	51.5 / 48.5 read through slab plastic · flagged low confidence	A\$5,929 VERIFIED SALES
Charizard δ Crystal Guardians #4	raw card no label present	54.2 / 45.8 · 51.0 / 49.0 passes PSA 10 · 59% confidence	A\$207 raw ESTIMATED

NOTE THE CONFIDENCE COLUMN

The three cards were read at 100%, low, and 59% confidence respectively. A card measured through slab plastic at an angle gives a less reliable reading than one photographed flat, and the system says so on the result rather than presenting every measurement as equally certain. **Knowing when a reading is soft is as important as the reading itself.** Note also that the Umbreon is the **#215 alternate art**, not the ordinary #95 from the same set — the two differ roughly thirtyfold in value, which is why the printing is settled before any price is fetched.

What it will not do

A valuation tool is only useful if its numbers can be relied on. These are the limits the system holds to, even when holding to them means showing less.

- ✗ **It will not grade a photograph it cannot read.** A clear reason and a re-shoot instruction are more useful than a number nobody should trust.
- ✗ **It will not let an image model set a price.** A model may propose a card's name, and that proposal is confirmed against a real catalogue. Prices come only from market feeds and completed sales.
- ✗ **It will not present an estimate as a sale.** Calculated figures are labelled everywhere they appear, including in the headline value.
- ✗ **It will not price a card it cannot confirm.** Where the exact printing is uncertain, no price is shown at all.
- ✗ **It will not guess at centering with nothing to measure.** Full-art cards have no printed border, and the system reports that rather than inventing a figure.
- ✗ **It will not leave a blank unexplained.** Wherever a value is missing, the interface states the reason on the card itself.

Working now, and what follows

WORKING TODAY

Grading label reading — company, grade and certificate number taken straight from the photograph

Centering — measured against the PSA standard with published confidence

Exact printing identification across eight card games

Graded valuation from completed sales, with the sample shown

Photo quality gate that declines rather than guesses

Australian dollars at daily central-bank rates, plus 29 other currencies

NEXT

Certificate lookup. Read the certificate number directly from the grading company, collapsing identification and grade into a single confirmed fact.

Trained condition models. Replace rule-based corner, edge and surface scoring with models trained against real graded outcomes.

Graded valuation beyond Pokémon. Extend slab pricing to Magic, One Piece and Lorcan.

Collection submission. Scan a full collection and rank it by what is worth grading first.

Identification and centering are production quality today. Condition sub-grades are a working first version and are labelled as estimates in the product. Valuation is real and sourced, and its precision improves as the depth of completed-sales data grows.